Graphene Supercapacitors

The Centre for Materials for Electronics Technology is a premier research centre in Electronic materials under Ministry of Electronics and Information Technology, Government of India. C-MET is engaged in the development of process/product/technology and transfer of the technology in the area of Electronic materials, components and devices to cater to India's strategic and industrial applications, exploiting indigenous resources of raw materials.

The C-MET family comprises three Centres, located at Pune, Hyderabad and Thrissur with specialized research mandate at each place. C-MET, Thrissur is concentrating on the development of electronic ceramic materials and devices. Its main areas of focus include Clean energy materials, supercapacitors, nano-materials, nano-porous materials, sensors and actuators, microwave materials & nano-structured electronics.

Supercapacitor can be used in a wide range of energy capture and storage applications and are used either by themselves as the primary power source or in combination with batteries or fuel cells. Some advantages of supercapacitors over more traditional energy storage devices include high power capability, long life, wide thermal operating range, low weight, flexible packaging, and low maintenance. Supercapacitors have already established their use in automobile sector for hybrid drives and regenerative braking systems. They have enormous opportunities in consumer electronics. Graphene is a promising material for supercapacitor electrodes owing to its low mass density, excellent electronic conductivity, and high surface area. Moreover, the intrinsic capacitance of graphene was recently found to be 21 $\mu F/cm^2$. In addition, graphene can be synthesized from the abundantly available and cost-effective graphite. Super capacitors based on these materials can have the cost and performance that would dramatically accelerate their adoption in a wide range of energy storage applications.

Main uses and domain

Automotive sector

One of the most promising fields of application for supercapacitors is in the automotive sector. Supercapacitors are being used for many potential applications both in electric vehicles and

conventional automobiles based on fossil fuels. Some of the applications explored globally for supercapacitor in the automotive sector are regenerative braking, Engine starting, Start-Stop systems, Short term back up power and peak power assist for high power functions.

Energy sector

More and more renewable energy sources such as solar energy, wind energy etc. are used for electrical power generation and hence electrical storage will be a requirement to level the load when the sun doesn't shine or the wind doesn't blow in a predictable manner. In addition to load leveling, grid storage will allow peak shaving where stored electricity from non peak demand periods can be used during peak load times.

Consumer Electronics

Supercapacitors have applications in multifunctional pocket appliances and can also cater the requirements for high pulse power application as in the case of digital cameras and wireless systems with burst-mode communications. Supercapacitors are also attractive for fast charge applications like instant charge hand tools like power drills.

ABOUT THE PRODUCTS:

C-MET has already developed a process for the production of graphene from natural graphite flakes through chemical route. We have developed graphene electrodes suitable for supercapacitors. C-MET has developed graphene based supercapacitors having capacitance of 1F to 200F.



Graphene supercapacitors developed by C-MET

C-MET has filed an Indian patent application (No. 293/CHE/2015) for this technology

Specifications of graphene Supercapacitors

1.	Capacitance	1F to 200F
2.	ESR	2 ohms to 6 milli ohms (Based on capacitance)
3.	Working Voltage	2.5 V

Technology Transfer

C-MET, Thrissur would like to invite competent industrial partners for commercialization of the above listed product through transfer of technology. The technology will be transferred on non-exclusive basis. The ToT package contains the following

1. Document for design and fabrication
2. Processing details
3. Product specification
4. Test plan and procedure
5. Training

The level of technology readiness at C-MET

The technology readiness level is 5

Period of training under the agreement

C-MET, Thrissur will give 2 months training to the engineers of the ToT partner at the premises of C-MET. Demonstration of 20 numbers of supercapacitors will be provided.

Who can apply?

Industries/ start ups can approach C-MET, Thrissur for the technology. Professionally managed Companies and Corporates are also welcome to apply for the technology. A Committee of experts constituted by the C-MET will assess capabilities and strengths of the industry before finalizing the technology partners.

How to apply?

Interested companies may send expression of interest with their details by filling the following questionnaire along with supporting documents to:

Programme Co-ordinator

Centre for Materials for Electronics Technology (C-MET)

Panchawati, Off Pashan Road, Pune:411008

Email: prog.coord@cmet.gov.in

General:

The industry willing to take technology for commercial exploitation will be required to enter into a ToT agreement with C-MET, Thrissur as per the terms and conditions approved by the competent authority in the MeitY prescribed format.

Note:

1. Optionally, for interested parties, technology for graphene through chemical route will be given on additional ToT fee.